Technical Requirements Document (TRD)

Abdur Rafey — Personal Portfolio Website

Version: 1.0 | **Author:** Abdur Rafey | **Date:** October 2026

1. Purpose & Scope

This document describes the architecture, technology choices, component design and deployment strategy for the personal portfolio website of Abdur Rafey — a Computer Science graduate from FAST-NUCES Lahore specialised in AI automation and agentic systems.

Scope: A fully static, client-rendered portfolio site with four routes (Home, O Levels, A Levels, FAST-NUCES). No backend, no database; all content is sourced from a single typed TypeScript file.

Live URL: `https://abdur-rafey-portfolio.vercel.app` *(auto-deployed on push to `main`)*

2. Tech Stack & Rationale

Layer	Technology	Rationale
Framework	**Next.js 14** (App Router, TypeScript)	Industry-standard React meta-framework with static export, optimised fonts, and SEO metadata APIs out of the box
Styling	**Tailwind CSS v3**	Utility-first, design-system-friendly, zero runtime CSS overhead
Animation	**Framer Motion v11**	Production-grade spring physics, `useInView`, layout animations; minimal bundle overhead
Icons	**react-icons v5**	Tree-shakeable SVG icon library (SI, FI sets) for tech and UI icons
Type Effects	**react-type-animation**	Lightweight hero typing effect
Intersection	**react-intersection-observer**	Scroll-reveal triggers for stat counters
Counters	**react-countup**	Animated number counters in the stats strip
Fonts	**next/font/google** (Inter + JetBrains Mono)	Zero-CLS font loading via Next.js font optimisation
Container	**Docker** (multi-stage) + **nginx:alpine**	Production-ready static serving with gzip and security headers
CI/CD	**GitHub Actions → Vercel**	Auto-deploy on every push to `main`; preview deployments for PRs

Static Export Strategy: `output: 'export'` in `next.config.js` compiles the site to the `out/` directory as flat HTML/JS/CSS — perfectly suited for nginx serving and CDN caching. Vercel handles the same output natively.

3. Route Map

Route	File	Description
`/`	`src/app/page.tsx`	Full home page: hero, stats, about, education snapshot, experience, projects, skills, contact
`/olevels`	`src/app/olevels/page.tsx` + `client.tsx`	Cambridge O Levels — subjects, grades, activities, societies
`/alevels`	`src/app/alevels/page.tsx` + `client.tsx`	Cambridge A Levels — subjects, grades, activities, societies
`/fast`	`src/app/fast/page.tsx` + `client.tsx`	FAST-NUCES BS CS — courses, CGPA, activities, societies

Server components (`page.tsx`) carry `metadata` exports for SEO; interactive logic is in `client.tsx` (`'use client'`).

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4. Component Architecture

``

```
src/
├── app/
│ ├── layout.tsx # Root layout: fonts, metadata, Navbar, Footer, CustomCursor, ParticleBackground
│ ├── globals.css # Design tokens, neon utilities, scrollbar, focus styles
│ ├── page.tsx # Home page (all sections)
│ ├── olevels/ # O Levels route
│ ├── alevels/ # A Levels route
│ └── fast/ # FAST-NUCES route
├── components/
│ ├── Navbar.tsx # Fixed top nav, active-route highlight, mobile menu
│ ├── Footer.tsx # Brand, nav links, socials, copyright
│ ├── CustomCursor.tsx # Neon glow cursor with spring animation
│ ├── ParticleBackground.tsx # Canvas particle system reacting to mouse
│ ├── Section.tsx # Scroll-reveal section wrapper + SectionHeading
│ ├── Card.tsx # 3D tilt card with dynamic glow on hover
│ ├── NeonIcon.tsx # Icon with wiggle, glow, tooltip on hover + click
│ ├── Timeline.tsx # Experience timeline with animated dot connectors
│ ├── SkillGrid.tsx # Category-grouped skill icons using NeonIcon
│ └── ProjectCard.tsx # Project card with tags, tilt, conditional links
```

```
└── data/
└── content.ts # Single source of truth — all portfolio content (typed)
```

Key design patterns:

- **Server/Client split:** Server components for metadata, client for interactivity. No unneeded 'use client' at layout level.
- **Single content file:** content.ts exports typed interfaces and all data. Editing the portfolio requires only editing this file.
- **Placeholder pattern:** Fields not yet provided by the owner are marked "[TO FILL]" and rendered with an orange-dashed badge so they are visually obvious.

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5. Content Model

All content is stored in src/data/content.ts and typed with TypeScript interfaces:

- profile — name, taglines, summary, email, links
- oLevels — institution, subjects (Mathematics A* confirmed), extracurricular, societies
- aLevels — institution, period, subjects (Math A, CS A, Physics A), extracurricular, IYLO leadership
- fast — degree, campus, period, CGPA, courses, leadership roles (FAST Orators' League, GLI, VOICE, IYLO), core skills developed
- experiences[] — role, company, location, period, bullet points, tags
- projects[] — title, description, tags, github, demo
- skillCategories[] — category name and array of { name, icon } skills
- stats[] — label, value, suffix (for animated counters)

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6. Docker / nginx Setup

Multi-Stage Build

- **Stage 1** (node:20-alpine): installs deps (npm ci), runs next build → outputs out/
- **Stage 2** (nginx:alpine): copies out/ into /usr/share/nginx/html; uses custom nginx.conf

nginx Features

- **gzip** compression for text, JS, CSS, SVG, fonts
- **Long-cache headers** (Cache-Control: public, immutable, 1 year) for all static assets
- **Security headers**: X-Frame-Options, X-Content-Type-Options, CSP, Permissions-Policy
- **Clean URLs** via try_files $uri $uri.html $uri/ /404.html
- **Custom 404** page

Running Locally

```bash
docker compose up --build
# → http://localhost:8080
```

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7. CI/CD Flow

```
Developer pushes to main
↓
GitHub repository (abdur-rafey-portfolio)
↓
Vercel detects push (GitHub integration)
↓
```

```
Vercel runs: npm install → next build → deploys out/

↓

https://abdur-rafey-portfolio.vercel.app (live, ~30s)
```

- **Pull Requests** get automatic preview URLs from Vercel.
- main branch is always deployable (CI gate: next build must pass).

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8. Performance & Accessibility Targets

Metric	Target
Lighthouse Performance	≥ 90
Lighthouse Accessibility	≥ 90
Lighthouse Best Practices	≥ 95
Lighthouse SEO	≥ 95

Techniques used:

- next/font/google for zero-CLS font loading
- images: { unoptimized: true } with static exports (images optimised manually)
- prefers-reduced-motion media query respected globally in CSS
- Semantic HTML: , , , , , single per page

- Keyboard focus states via :focus-visible with neon outline
- aria-label on all icon-only links, aria-current="page" on active nav links
- alt text on all images

9. AI Tools Used

- **Google Antigravity (AGY)** — primary agent that scaffolded the entire project, wrote all components, set up Docker/nginx, and handled Git commits
- **Claude Sonnet (via Antigravity)** — code generation and review

10. Future Improvements

- Add a blog/` route powered by MDX for technical writing
- Integrate a CMS (e.g. Sanity, Contentlayer) so content can be edited without touching code
- Add contact form with email delivery (Resend / Nodemailer)
- Dark/light theme toggle
- Internationalise (Urdu language support)
- Add Cypress E2E tests and Lighthouse CI GitHub Action